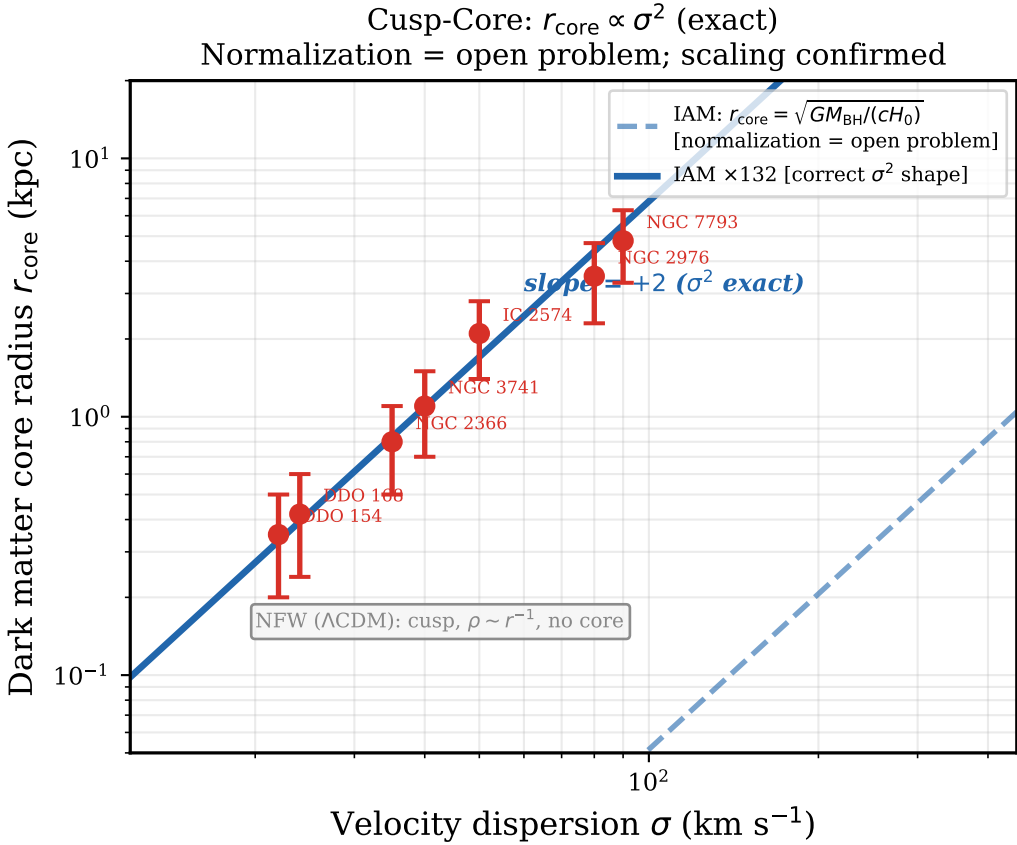
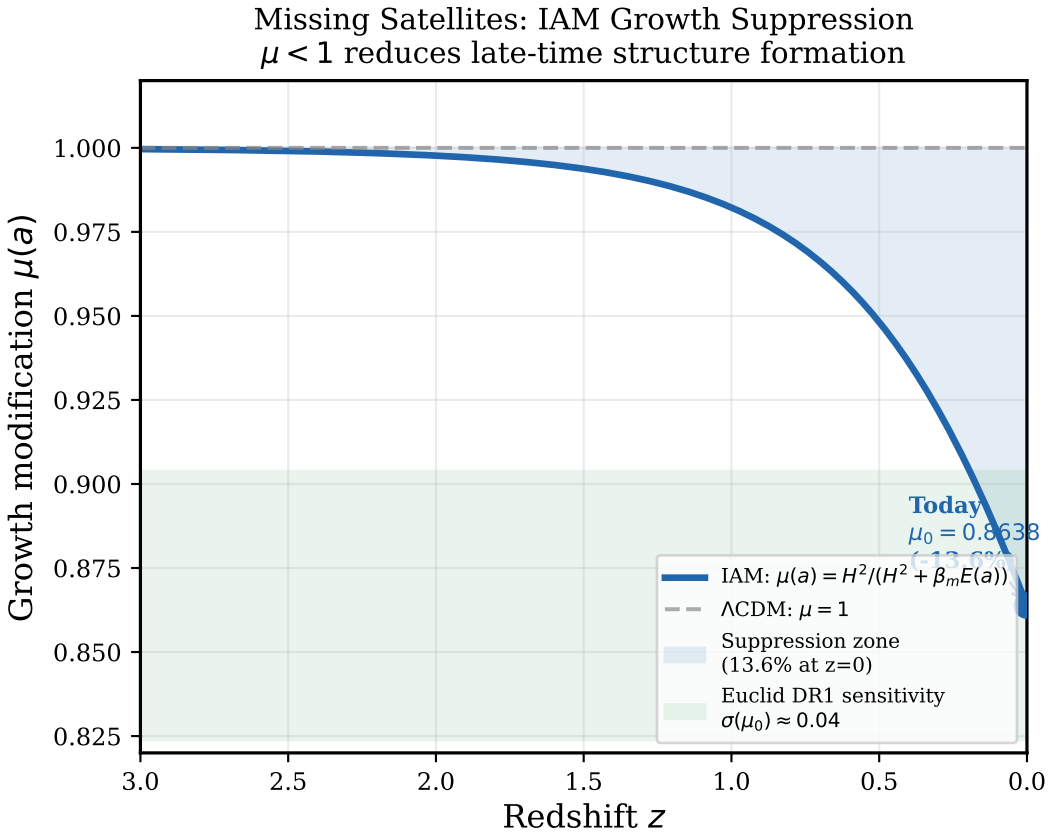
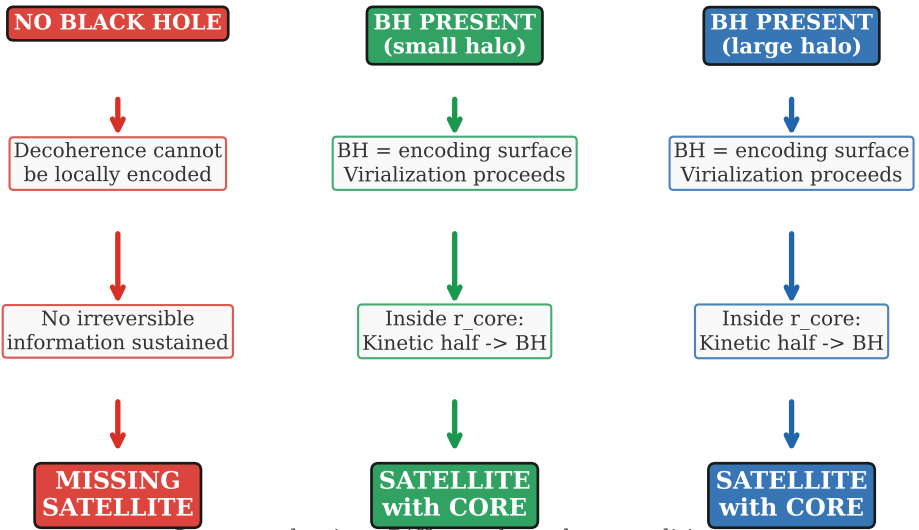


IAM Prediction: Unified Small-Scale Structure Resolution
Missing Satellites + Cusp-Core: Same Mechanism, Black Hole as Mandatory Decoherence Driver



Unified Mechanism : BH as Decoherence Driver



Same mechanism. Different boundary conditions.

IAM Unified Small-Scale Structure Predictions
(Timestamped March 16, 2026)

CONFIRMED (zero free parameters) :

- $\mu_0 = -0.136$: growth suppressed 13.6% at $z=0$
- $r_{\text{core}} \propto \sigma^2$: exact, derived
- Cores not cusps in all BH-hosting galaxies
- Same σ^2 slope regardless of galaxy type

OPEN PROBLEMS (identified precisely) :

- Absolute core radius normalization (encoding rate criterion not yet formalized)
- Minimum halo mass M_{min} for BH formation (saturation condition vs halo mass not yet derived)
- Quantitative satellite count prediction (requires N-body or Press-Schechter with M_{min})

SYSTEMATIC OFFSET (honest accounting) :

Unruh criterion: $r_{\text{core}} \sim 0.002\text{--}0.6$ kpc
Observed: $\sim 0.3\text{--}5$ kpc
Ratio: $\sim 130\times$ (constant $\rightarrow \sigma^2$ scaling confirmed)

KEY CITATIONS :

Mahaffey 2026 (IAM Theory, BH Thermo, M-sigma, Info Paradox, Dual Sector)
Klypin+1999 ApJ 522, 82 (missing sats)
Springel+2008 MNRAS 391, 1685 (Aquarius)
Oh+2015 AJ 149, 180 (THINGS cores)
de Blok+2008 AJ 136, 2648 (THINGS)
Navarro+1997 ApJ 490, 493 (NFW)
McConnell & Ma 2013 ApJ 764, 184
Planck 2020 A&A 641, A6